

Forecast one series with ARIMA

20 minutes of ideas. 40 minutes of hands-on.

CCIS faculty workshop | Oct 2, 2026 | 3 PM to 4 PM

Slides, notebooks, and web explorer: github.com/ahleksu/arima-workshop-ccis

The hour

Minute	What happens
0 to 20	Lecture in seven short beats
20 to 22	Open notebook 01 in Colab
22 to 33	Block A. Stationarity (01)
33 to 45	Block B. AR, MA, ACF, PACF (02)
45 to 57	Block C. Fit and forecast an ARIMA (03)
57 to 60	Wrap-up and take-home

Beat 1. The question

What comes next?

- A time series is a list of values in time order
- Yesterday helps to predict today, so the order carries information
- ARIMA learns that link from the past and projects it forward
- It is also the **baseline** that a newer model must beat

Name one series from your own work.

When ARIMA fits, and when it does not

Good fit

- One series, regular spacing
- 50 to a few thousand points
- Patterns you can describe: trend, season, memory
- You need intervals and a model you can explain

Poor fit

- Many related series, or huge feature sets
- Strong non-linear or regime-switching behavior
- Very irregular or event-driven data
- Very long seasons (365 with daily data)

Beat 2. Look first

Four parts of a series:

- **Trend:** a slow rise or fall
- **Seasonality:** a pattern that repeats at a fixed period
- **Cycle:** long swings with no fixed period
- **Noise:** the part that nothing explains

Our data is daily electricity demand. It is **synthetic**, so we know the truth.

Plot first. Test second.

Beat 3. Stationarity

A series is **stationary** when its rules do not change over time:

- the mean stays constant
- the spread stays constant
- the link between a value and the value k steps earlier depends only on k

Why: ARIMA learns **one** set of coefficients. That only works if the pattern holds in the future.

A trend breaks it.

Differencing and the ADF test

Difference: replace each value by its change, $y_t - y_{t-1}$. A rising line becomes a flat line.

ADF test

- Null hypothesis: the series is **non-stationary** (unit root)
- **Small** p-value (below 0.05): the series looks stationary

Our data: level $p = 0.63$, first difference p near 0.

Difference as little as possible. Each extra difference adds noise.

Beat 4. Two kinds of memory

AR(p): memory of **values**

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

MA(q): memory of **shocks**

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

The error ε_t is the surprise: the part that no earlier value predicts.

Reading the fingerprints

Process	ACF	PACF
AR(p)	Fades	Stops after lag p
MA(q)	Stops after lag q	Fades
Both	Fades	Fades
White noise	No spikes	No spikes

The plot that stops names the order.

The blue band is about $\pm 1.96 / \sqrt{n}$. One spike in twenty crosses it by chance.

Beat 5. Three dials

ARIMA(p, d, q)

- **p**: how many past values
 - **d**: how many differences
 - **q**: how many past surprises
1. Choose **d**: difference until stationary, and no more
 2. Read the ACF and PACF for candidate **p** and **q**
 3. Fit the candidates and compare **AIC** and **BIC** (lower is better)
 4. Within about 2 points, take the **simpler** model

Our result (notebook 03)

- Weekly demand, 182 training weeks
- ADF says $d = 1$
- Lowest AIC: ARIMA(2,1,1). Lowest BIC: ARIMA(0,1,1)
- Five models are within 2 AIC points, so the **parsimony rule** picks **ARIMA(0,1,1)**

You will build this yourself in Block C.

Beat 6. Can you trust it?

1. **Residuals** (what the model missed) must look like white noise
2. **Ljung-Box test**: a **large** p-value is good here. This is the opposite of ADF.
3. **Hold-out test**: fit on the past, forecast the last part, compare
4. **Baseline**: beat a naive forecast, or the model has earned nothing

Our test: 26 weeks held out. Error 18.5 GWh, 1.1 percent of the mean.

A low AIC does not prove that the model is adequate.

Beat 7. Handoff

For the next 40 minutes:

1. Open the link for notebook `01` on the Hands-on page
2. Run the cells **top to bottom** (Shift + Enter)
3. Read the text above each cell before you run it
4. Answer the check question at the end of each block
5. Raise a hand when a cell fails

Block A. Stationarity (notebook 01)

11 minutes. Run sections 1 to 5. Skip section 6 (Box-Cox).

Check question

What is the ADF p-value of the level, and of the first difference?

Block B. AR, MA, ACF, PACF (notebook 02)

12 minutes. Run sections 1 to 3, then the section 6 exercise. Skip sections 4 and 5.

Check question

For series A, do AIC and BIC find the true model?

Block C. Fit and forecast (notebook 03)

12 minutes. Run sections 1 to 5, 7, and 8. Skip section 6 (likelihood plot).

Check question

Which order does the parsimony rule pick, and what is the test error?

Wrap-up

- Plot first, test second
- Difference as little as possible
- The plot that stops names the order
- Fit, then diagnose, then decide
- A forecast needs a baseline and a hold-out test

After today: notebook 04 (seasonal terms and outside drivers), 05 (validation), 06 (your own data).

Reviewing a forecasting chapter: 8 questions

1. Is the data source, frequency, and period stated?
2. Were breaks and outliers found and explained?
3. How was stationarity decided, and which tests?
4. How were orders chosen? Is the grid stated?
5. Do the residual checks pass?
6. Is there a baseline, and does the model beat it?
7. Is the evaluation out of sample, from many origins?
8. Are intervals reported, with their coverage?

Your next hour

1. Open **notebook 06**. Run it once on the sample.
2. Change `DATA_PATH` and four settings. Run it on your series.
3. Read the data checklist output before you trust anything.
4. Copy the methods paragraph and edit it.

Privacy: keep sensitive data off Colab. Use the local setup script.

Where to go next

- *Forecasting: Principles and Practice* (3rd edition), free online: otexts.com/fpp3
- statsmodels time series docs: statsmodels.org/stable/tsa.html
- `pmdarima` docs: alkaline-ml.com/pmdarima
- Extensions: ARIMAX with lagged drivers, state-space models, ETS, hierarchical forecasting, probabilistic metrics

Everything from today: github.com/ahleksu/arima-workshop-ccis

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Questions

Come back to the series you wrote on the board at the start.